Physics and Mathematics Reference
Recreated Heat Transfer Study Model

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1 Purpose and scope

This document describes the physics and mathematics implemented in the recreated study script `heat_transfer_study_recreated.py`, including:

- legacy winter heating from the exported heater/aeration sweep;
- simplified winter wall-blanket heating;
- legacy summer spot-cooler plus assist-blower cooling;
- simplified summer single vertical frozen-bottle column cooling;
- shared pressure-drop, moisture-loss, two-node bed, radial plume, and year-round layers.

Primary implementation anchors:

- runtime dispatch and mode selection: `heat_transfer_study_recreated.py:L9427-L9557`;
- heating mode config: `heat_transfer_study_recreated.py:L6500-L6520`;
- cooling mode config: `heat_transfer_study_recreated.py:L6521-L6633`;
- optimization config and temperature-goal toggles: `heat_transfer_study_recreated.py:L2506-L2586`.

2 Execution modes and configuration controls

Two run layers can be enabled independently via `run_modes.heating` and `run_modes.cooling`. The recreated profiles are:

- legacy+ profile: `study_config_legacy_plus.json:L8-L399`;
- simplified vertical/wall-blanket profile: `study_config_simplified_vertical.json:L8-L446`.

Mode-selecting fields:

- `heating_mode.method`: `legacy_export_from_matlab` or `wall_blanket_conduction` (`study_config_simplified_vertical.json:L8-L23`);
- `cooling_mode.method`: `spot_cooler_with_assist_blower` or `vertical_frozen_bottle_column` (`study_config_simplified_vertical.json:L236-L239`).

Optimization and seasonal-goal toggles:

- `optimization.method` and `cooling_mode.optimization.method`: `nsga2` or `differential_evolution`;
- `temperature_goal_mode`: `legacy_extreme_node` or `seasonal_threshold`;

- threshold targets: `winter_minimum_target_C`, `summer_maximum_target_C`.

Config anchors: `study_config_legacy_plus.json:L264-L330`, `study_config_simplified_vertical.json:L311-L377`.

3 Thermophysical and psychrometric closures

Air properties (ideal-gas and Sutherland-law transport):

$$\rho = \frac{p}{R_a T}, \tag{1}$$

$$\mu = \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \frac{T_0 + S}{T + S}, \tag{2}$$

$$k = \frac{\mu c_p}{Pr}. \tag{3}$$

Implemented in `heat_transfer_study_recreated.py:L4972-L4980`.

Humidity-ratio relations:

$$w = 0.62198 \, \frac{p_v}{p - p_v}, \tag{4}$$

$$p_v = \phi \, p_{\mathrm{sat}}(T). \tag{5}$$

Implemented in `heat_transfer_study_recreated.py:L5016-L5029`, with dew-point inversion in `heat_transfer_study_recreated.py:L5047-L5060`. Water saturation pressure and latent heat use the imported IAPWS backend (`heat_transfer_study_recreated.py:L4983-L5014`).

4 Convection and friction correlations

Wire-side external forced convection uses Churchill–Bernstein:

$$Nu = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{\left(1 + (0.4/Pr)^{2/3}\right)^{1/4}} \left(1 + \left(\frac{Re}{282000}\right)^{5/8}\right)^{4/5}. \tag{6}$$

Implemented in `heat_transfer_study_recreated.py:L5150-L5157`.

Tube friction factor uses Churchill's smooth-pipe form:

$$f = 8 \left[\left(\frac{8}{Re}\right)^{12} + \frac{1}{(A + B)^{3/2}}\right]^{1/12}, \tag{7}$$

with laminar fallback $f = 64/Re$, implemented in `heat_transfer_study_recreated.py:L5159-L5166`.

Internal tube Nusselt closure:

- laminar developing flow: $Nu = 3.66 + \frac{0.0668\, Gz}{1 + 0.04\, Gz^{2/3}}$, $Gz = Re\, Pr\, D/L$;
- turbulent flow: Gnielinski-type relation using f.

Implemented in `heat_transfer_study_recreated.py:L5168-L5177`.

Natural convection for wall-loss closure:

- vertical surfaces: Churchill–Chu style $Nu(Ra, Pr)$;
- horizontal surfaces: stable/unstable orientation piecewise $Nu(Ra)$.

Implemented in `heat_transfer_study_recreated.py:L5062-L5099`.

5 Overall U and wall-loss resistances

The cylindrical overall U on the outer-area basis combines inner convection, wall conduction, optional insulation conduction, and outer convection:

$$\frac{1}{U_o} = \frac{D_o}{D_i h_i} + \frac{D_o \ln(D_w/D_i)}{2k_w} + \frac{D_o \ln(D_o/D_w)}{2k_{ins}} + \frac{1}{h_o}. \tag{8}$$

Implemented in `heat_transfer_study_recreated.py:L5179-L5215`.

6 Legacy winter heating path (exported heater/aeration design points)

For legacy heating, each sampled design point carries electrical topology, heater-tube segment results, aeration-tube release, pressure loss, and bed-node temperatures via `simulate_heating_reference_point` (`heat_transfer_study_recreated.py:L5521-L6283`).

Electrical grouping uses strings of series tubes under one supply voltage. For one string:

$$I_{\text{string}} = \frac{V}{n_s R_{\text{tube}}}, \tag{9}$$

$$P_{\text{tube}} = I_{\text{string}}^2 R_{\text{tube}}. \tag{10}$$

The full point aggregates all strings/tubes.

Heater-tube segment energy update:

$$\dot{q}_{\text{gen},i} = P_{\text{tube}} \frac{\Delta x}{L}, \tag{11}$$

$$\dot{q}_{\text{loss},i} = U_{\text{loss}} P_{\text{exp}} \Delta x \max(T_{air,i} - T_\infty, 0), \tag{12}$$

$$T_{air,i+1} = T_{air,i} + \frac{\dot{q}_{\text{gen},i} - \dot{q}_{\text{loss},i}}{\dot{m} c_p}. \tag{13}$$

Implemented in `heat_transfer_study_recreated.py:L5248-L5347`.

7 Simplified winter wall-blanket conduction path

When `heating_mode.method=wall_blanket_conduction`, electrical aeration heating is replaced by wall blankets under insulation (`heat_transfer_study_recreated.py:L5523-L5630`). No aeration-driven heating or nichrome heater-tube heating is used in this winter branch.

Configured blanket hardware basis (per blanket):

- dimensions $0.250 \times 1.740 \times 0.0015$ m (10 in $\times$ 70 in $\times$ 1.5 mm);
- rated electrical power 1200 W at 110 V;

- modeled max blanket surface temperature 120 °C.

Blanket surface temperature is duty-ramped:

$$T_{blanket} = T_\infty + \delta\,(T_{blanket,\max} - T_\infty), \qquad 0 \le \delta \le 1. \tag{14}$$

Inward and outward heat fluxes:

$$q''_{\text{in}} = \frac{\max(T_{blanket} - T_{bed}, 0)}{R_{\text{in}}}, \tag{15}$$

$$q''_{\text{out}} = \frac{\max(T_{blanket} - T_\infty, 0)}{R_{\text{out}}}. \tag{16}$$

Total powers:

$$Q_{\text{to bed}} = N_b A_b q''_{\text{in}}\,\eta_{coupling}, \tag{17}$$

$$Q_{\text{to amb}} = N_b A_b q''_{\text{out}}, \tag{18}$$

$$P_{\text{elec}} = Q_{\text{to bed}} + Q_{\text{to amb}}, \tag{19}$$

$$I_{\text{total}} = \frac{P_{\text{elec}}}{V_{supply}}. \tag{20}$$

Thermal response still uses the same two-node bed network and the same wall-loss model as all other branches.

8 Summer inlet-conditioning models

8.1 Spot-cooler plus assist-blower mode

For `spot_cooler_with_assist_blower`, requested sensible cooling is

$$Q_{\text{req}} = \dot{m}c_p(T_{amb} - T_{target})_+. \tag{21}$$

Actual load is capped by configured/rated sensible capacity, then

$$T_{supply} = T_{amb} - \frac{Q_{\text{actual}}}{\dot{m}c_p}. \tag{22}$$

Condensate is estimated from psychrometric carrying-capacity difference with an optional rated dehumidification cap. Implemented in `heat_transfer_study_recreated.py:L3512-L3598`.

8.2 Simplified vertical frozen-bottle column mode

For `vertical_frozen_bottle_column`, a single central vertical tube is used; fan flow is downward through packed frozen bottles (`heat_transfer_study_recreated.py:L3601-L3699`, `heat_transfer_study_recreated.py:L3703-L4108`).

Column superficial velocity:

$$u_s = \frac{\dot{V}}{A_{col}}, \qquad A_{col} = \frac{\pi D_{col}^2}{4}. \tag{23}$$

Convective coefficient and effective UA:

$$h_{col} = h_0 + C_v u_s^{0.8}, \tag{24}$$

$$UA_{col} = h_{col} A_{exp} \quad \text{(or fixed override if configured).} \tag{25}$$

Unconstrained outlet from an exponential UA model:

$$T_{out,ideal} = T_{sink} + (T_{amb} - T_{sink}) \exp\left(-\frac{UA_{col}}{\dot{m} c_p}\right). \tag{26}$$

Frozen-bottle capacity cap:

$$E_{ice} = m_w [c_{p,ice}(0 - T_{ice,0}) + h_{fus} + c_{p,liq}(T_{post} - 0)], \tag{27}$$

$$\dot{Q}_{cap} = \frac{E_{ice}}{\tau_{replace}}, \tag{28}$$

$$\dot{Q}_{actual} = \min(\dot{Q}_{ideal}, \dot{Q}_{cap}). \tag{29}$$

The resulting T_{supply}, humidity, and condensate then feed the same aeration/bed solvers used by the other summer branch.

9 Aeration pressure model and closed-end perforated-tube release

Both heating and cooling branches use the same core release and pressure closures.

Per-segment perforation release:

$$\dot{m}_{\mathrm{rel},i} = C_d A_{h,i} \sqrt{2\rho_i \left(p_{tube,i} - p_{bed,i}\right)_+}. \tag{30}$$

Implemented in `heat_transfer_study_recreated.py:L3154-L3214`, `heat_transfer_study_recreated.py:L3366-L3436`.

Segmentwise bed backpressure uses an Ergun pressure gradient:

$$\frac{dp}{dx} = 150 \frac{(1-\epsilon)^2}{\epsilon^3} \frac{\mu u_s}{d_p^2} + 1.75 \frac{1-\epsilon}{\epsilon^3} \frac{\rho u_s^2}{d_p}, \tag{31}$$

with local path length to open top or vent. Implemented in `heat_transfer_study_recreated.py:L3216-L3311`, `heat_transfer_study_recreated.py:L3334-L3364`.

Closed-end pressure is solved by scalar shooting:

$$\dot{m}_{\mathrm{rem},i+1} = \dot{m}_{\mathrm{rem},i} - \dot{m}_{\mathrm{rel},i}, \tag{32}$$

$$p_{i+1} = \max(p_i - \Delta p_{fric,i},\, p_{bed,i}). \tag{33}$$

The inlet gauge pressure is bisected until terminal residual flow is below tolerance. Implemented in `heat_transfer_study_recreated.py:L3442-L3510`.

10 Distribution-network losses

Legacy manifold-network losses include:

- header entry and header-to-splitter contraction;
- splitter body/tee/valve surrogate losses (Hooper 2-K style);
- branch connector friction;
- connector-to-branch contraction/expansion losses.

Implemented in `heat_transfer_study_recreated.py:L5787-L6323`.

In simplified vertical-column summer mode, these can be bypassed via `vertical_column.bypassDistributionNetwork=true` (`heat_transfer_study_recreated.py:L3801-L3817`).

11 Aeration segment energy, jet transfer, and evaporation

Each aeration segment contributes:

$$Q_{\text{to bed},i} = Q_{\text{wall},i} + Q_{\text{jet},i} - Q_{\text{evap},i}, \tag{34}$$

$$Q_{\text{jet},i} = \dot{m}_{\text{rel},i} c_p (T_{\text{rel},i} - T_{bed}). \tag{35}$$

Evaporation rate is triple-limited:

$$\dot{m}_{evap,i} = \min(\dot{m}_{diff,i}, \dot{m}_{capacity,i}, \dot{m}_{heat,i}), \qquad Q_{\text{evap},i} = \dot{m}_{evap,i} h_{fg}. \tag{36}$$

Implemented in `heat_transfer_study_recreated.py:L3160-L3200`, `heat_transfer_study_recreated.py:L5406-L5486`, `heat_transfer_study_recreated.py:L3892-L3999`.

12 Bin wall losses and two-node bed closure

The bin loss model assembles side, bottom, and top UA based on top state (open, covered+vent, covered) and bottom boundary mode: `heat_transfer_study_recreated.py:L6324-L6464`.

Lumped requirement:

$$Q_{req} = UA_{tot}(T_{set} - T_{amb}). \tag{37}$$

Implemented in `heat_transfer_study_recreated.py:L6466-L6486`.

Two-node steady state:

$$\begin{bmatrix} UA_b + UA_{12} & -UA_{12} \\ -UA_{12} & UA_t + UA_{12} \end{bmatrix} \begin{bmatrix} T_b \\ T_t \end{bmatrix} = \begin{bmatrix} Q_b + UA_b T_{amb} \\ Q_t + UA_t T_{amb} \end{bmatrix}, \tag{38}$$

with $Q_b = f_b Q_{bed}$, $Q_t = (1 - f_b) Q_{bed}$. Implemented in `heat_transfer_study_recreated.py:L6488-L6498`.

13 Radial plume and habitat exclusion post-processing

The post-processing plume uses release-air cooling away from the tube with configurable linear plume-radius spread:

$$r_p(x) = \min(r_{max},\, r_0 + \alpha x). \tag{39}$$

Sensible and latent sink terms are applied along radius; the "safe distance" is the first radius where $T(x) \leq T_{threshold}$. Implemented in `heat_transfer_study_recreated.py:L1365-L1708`.

Tube-layout and exclusion-area mapping is then computed on a 2-D cross-section grid using either explicit centers, custom single row, or rule-based patterns: `heat_transfer_study_recreated.py:L1709-L2050`.

14 Constraint residuals and objective metrics

14.1 Heating metrics

Heating residuals include current, wire temperature, outlet-air temperature, pressure drop, moisture loss, hole velocity, bottom/top min and max limits, spread limit, and heat shortfall:

$$r_k(x) = \max(0, g_k(x)). \tag{40}$$

Implemented in `heat_transfer_study_recreated.py:L2610-L2724`.

Legacy heating objective family:

- `power_W`;
- `minus_min_bed_temp_C`;
- `temp_spread_C`.

Threshold heating objective family:

- `winter_threshold_shortfall_C`;
- `power_W`;
- `temp_spread_C`.

14.2 Cooling metrics

Cooling residuals include assist-blower margin, pressure drop, moisture loss, hole velocity, temperature bounds/spread, and nonpositive cooling delivery. Implemented in `heat_transfer_study_recreated.py:L4112-L4225`.

Legacy cooling objective family:

- `max_bed_temp_C`;
- `mean_bed_temp_C`;
- `combined_cooling_power_W`.

Threshold cooling objective family:

- `summer_threshold_excess_C`;
- `combined_cooling_power_W`;
- `totalFlow_Lpm`.

15 Optimization methods

15.1 NSGA-II mode

When `method=nsga2`, constrained NSGA-II is applied directly on the candidate design-point set: `heat_transfer_study_recreated.py:L2889-L2919`, `heat_transfer_study_recreated.py:L4317-L4347`.

15.2 Differential-evolution mode

When `method=differential_evolution`, the solver runs multiple weighted scalarizations and then keeps the feasible non-dominated set.

For one weight vector w_j:

$$J_j(x) = \sum_i w_{j,i} \frac{f_i(x) - f_{i,\min}}{\Delta f_i} \; + \; \lambda \sum_k r_k(x), \tag{41}$$

where $r_k(x)$ are nonnegative constraint residuals.

Heating DE implementation: `heat_transfer_study_recreated.py:L2921-L3001`. Cooling DE implementation: `heat_transfer_study_recreated.py:L4349-L4424`.

16 Seasonal threshold objective mode

With `temperature_goal_mode=seasonal_threshold`:

- winter heating objective/priority includes `winter_threshold_shortfall_C` relative to `winter_minimum_target_C`;
- summer cooling objective/priority includes `summer_threshold_excess_C` relative to `summer_maximum_target_C`.

Implementation anchors: `heat_transfer_study_recreated.py:L2556-L2586`, `heat_transfer_study_recreated.py:L2633-L2645`, `heat_transfer_study_recreated.py:L4138-L4151`.

17 Year-round layer

The annual model builds a representative climate year via periodic Gaussian-kernel regression from monthly means/standard deviations, then applies freeze-day and heat-wave overrides: `heat_transfer_study_recreated.py:L6910-L7023`.

Daily simulation:

1. compute daily required heating/cooling from the same lumped UA model;
2. re-simulate selected heating or cooling reference point at that day's ambient;
3. apply duty, unmet-load accounting, electricity use, and water use.

Implemented in `heat_transfer_study_recreated.py:L7025-L7133` and `heat_transfer_study_recreated.py:L7134-L7265`.

18 Symbol glossary

Symbol	Meaning	Units
Q_{bed}	Net heat delivered to bed (positive heating, negative cooling)	W
Q_{req}	Lumped required load at setpoint and ambient	W
UA	Overall heat-loss conductance	W/K
T_b, T_t	Bottom and top bed-node temperatures	°C
$\dot{m}_{\mathrm{rel},i}$	Segment release mass flow from perforations	kg/s
Δp	Pressure drop / required static pressure	Pa
ϵ	Bed porosity in Ergun closure	–
d_p	Representative bed particle diameter	m
h_{fg}	Latent heat of vaporization	J/kg
$\dot{m}_{evap}$	Evaporation mass flow	kg/s
u_s	Superficial velocity (context-dependent)	m/s
δ	Wall-blanket duty fraction	–

19 Implementation files

- model script: `S:\T1\textbackslashAcademicResources\T1\textbackslashUCONN\T1\textbackslash10.Spring2026\T1\textbackslashCHEG4143W\T1\textbackslashProject\T1\textbackslashVermiHeating\T1\textbackslashHeatTransferStudy\T1\textbackslashReduced\T1\textbackslashheat_transfer_study_recreated.py`
- legacy-plus config: `S:\T1\textbackslashAcademicResources\T1\textbackslashUCONN\T1\textbackslash10.Spring2026\T1\textbackslashCHEG4143W\T1\textbackslashProject\T1\textbackslashVermiHeating\T1\textbackslashHeatTransferStudy\T1\textbackslashReduced\T1\textbackslashstudy_config_legacy_plus.json`
- simplified config: `S:\T1\textbackslashAcademicResources\T1\textbackslashUCONN\T1\textbackslash10.Spring2026\T1\textbackslashCHEG4143W\T1\textbackslashProject\T1\textbackslashVermiHeating\T1\textbackslashHeatTransferStudy\T1\textbackslashReduced\T1\textbackslashstudy_config_simplified_vertical.json`